Improving k-Nearest Neighbors on Fashion-MNIST with Learned Distances

From 85.2 % to 89.5 % test accuracy

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Team project with Akshar Nath Gorain, Sourodyuti Biswas Sanyal, Aditi Barman and Sanjana Mahata.
Code: github.com/workforakng/fashion-mnist-knn-optimization
Demo notebook: open in Google Colab

Abstract

The k-nearest neighbors (KNN) classifier labels an image by a vote among the most similar training images. On Fashion-MNIST, a dataset of clothing images, plain KNN on raw pixels reaches 85.24 % test accuracy. We study how far this can be raised while the final decision still comes from distances to stored training images. Principal component analysis (PCA) and a tuned KNN give 86.43 %. We then learn the distance with Soft-KNN, a smooth version of KNN that is close to neighborhood components analysis. A linear embedding gives 87.10 % and a small convolutional network gives 87.63 %. The final fusion model (E25) reaches 89.51 %. It is not plain KNN. It mixes a learned neural embedding with neighbor retrieval, ensembling, knowledge distillation and test-time augmentation. On this test set, single-run differences below about 0.6 percentage points are within noise. We also report changes that did not help and the limits of our protocol.

1 Introduction

The k-nearest neighbors (KNN) rule [1] is one of the simplest classifiers. It has no training step. To label a new image, it finds the k closest training images and takes a majority vote over their labels.

KNN has two problems on images. First, a 28×28 image has $D = 784$ pixels. In so many dimensions, the nearest and the farthest images can be at similar distances [2]. Second, flattening an image into a list of pixels loses which pixels are next to each other.

Fashion-MNIST [3] has 70,000 grayscale images of size 28×28. They show ten classes of clothing: T-shirt/top, Trouser, Pullover, Dress, Coat, Sandal, Shirt, Sneaker, Bag and Ankle boot. There are 60,000 training images and 10,000 test images. Some classes look alike, such as Shirt, Pullover and Coat.

Research question. How much can we improve KNN on Fashion-MNIST by reducing the input dimension, by learning the distance, and by combining several models?

Approach. Figure 1 shows the parts of the final system. We first tune classical KNN on principal component analysis (PCA) features. We then learn an embedding with Soft-KNN, a smooth version of KNN that can be trained by gradient descent. A small convolutional neural network (CNN) gives the best single embedding. We average three such networks and distill them into one network. The final fusion model (E25) is a weighted mix of three neighbor-based classifiers.

Contributions.

- A study of classical KNN on Fashion-MNIST: distance measures, PCA, the choice of k and query time (Section 4).
- A Soft-KNN model that learns the distance over a memory bank of training images. It follows neighborhood components analysis [4] and its scalable version [5]. We do not claim it is new (Section 5).
- A convolutional embedding, a seed ensemble, distillation and a weighted fusion that reach 89.51 % test accuracy (Section 6).
- A list of changes that did not help (Section 7.2) and the limits of our protocol (Section 8).

Every run is logged in a JSON (JavaScript Object Notation) file in the repository (the *ledger*, `results/results.json`). We name runs by their ledger ID, such as E1 or E25. Section 7.1 summarizes the main runs.

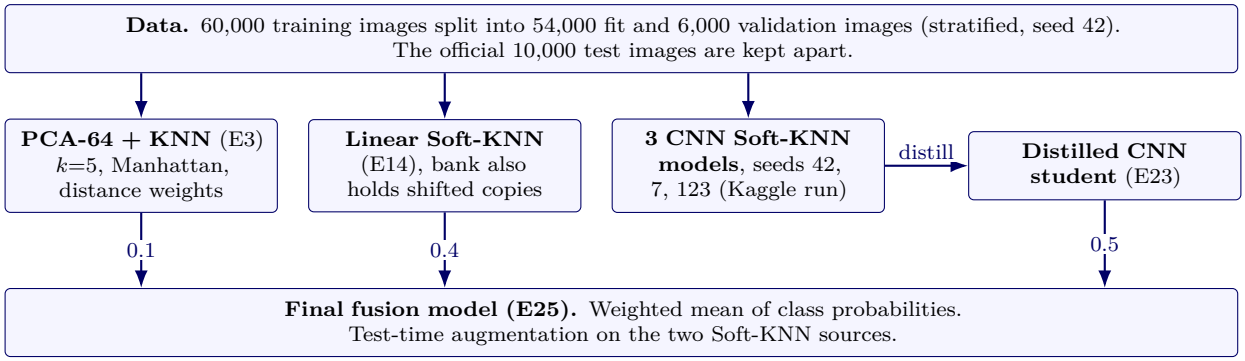


Figure 1: Parts of the final fusion model (E25). The numbers on the arrows are the fusion weights chosen on validation data. The three seed models feed the fusion only through distillation. Four other candidate sources received weight 0 (Section 6.4).

2 Related Work

Cover and Hart [1] showed that the error of the nearest-neighbor rule is at most twice the best possible error, given unlimited data. Beyer et al. [2] showed that in high dimensions, distances to the nearest and the farthest point can become nearly equal.

Several methods learn a distance for KNN. Neighborhood components analysis (NCA) [4] learns a linear map. It trains a soft version of the leave-one-out KNN error over all training points. Large margin nearest neighbor (LMNN) [6] learns a distance that keeps same-class neighbors close and pushes other classes away.

Wu et al. [7] store the embeddings of all training images in a *memory bank*. The bank is updated during training, so the network does not recompute every embedding at every step. Wu, Efros and Yu [5] combine such a bank with an NCA loss and a deep network, and they classify by KNN. Our Soft-KNN is close to this scalable NCA.

Knowledge distillation [8] trains a *student* network to match the class probabilities of a *teacher* network. We tune classical KNN with Optuna [9], a library for hyperparameter search.

3 Data and Protocol

Splits. We split the 60,000 training images into 54,000 *fit* images and 6,000 *validation* images. The split keeps the class balance (stratified, seed 42). The official 10,000 test images are used only to report results. Pixels are scaled to $[0, 1]$.

Rules. We choose models and settings by validation accuracy. We compute test accuracy after each choice, but no setting is chosen by test accuracy. The global seed is 42, except in the seed study.

Uncertainty. We write differences in percentage points (pp). On 10,000 test images, one binomial standard error (SE) of accuracy is about ± 0.3 pp. So a gap under about 0.6 pp between two single runs is not reliable. We mark such gaps as "within noise". On the 6,000 validation images, one SE is about 0.4 pp. We ran no significance tests.

Validation reuse. We used the same 6,000 validation images for many choices. These include the PCA dimension, the Optuna search, early stopping, the inference settings and the fusion weights. So the best validation accuracy (90.85 %) is optimistic. The test accuracy is the fair estimate.

Where the runs ran. The stages in `src/experiments.py` ran on the authors' local machine. The distilled student (E23) and its three teacher models were trained on Kaggle graphics processing units (GPUs) (`results/kaggle_*.json`). The seed models in E21 come from a separate run of the same seed script (`results/conv_seeds.json`).

4 Classical KNN Baselines

4.1 Distance Measures on Raw Pixels

For two images $x, x' \in [0, 1]^D$, we compare three distances:

$$\delta_2(x, x') = \|x - x'\|_2, \qquad \delta_1(x, x') = \textstyle\sum_{j=1}^{D} |x_j - x'_j|, \qquad \delta_{\cos}(x, x') = 1 - \frac{x^\top x'}{\|x\|_2 \, \|x'\|_2}. \tag{1}$$

Here δ_2 is the Euclidean distance, δ_1 the Manhattan distance and $\delta_{\cos}$ the cosine distance. KNN predicts $\hat{y}(x) = \arg\max_c \sum_{i \in \mathcal{N}_k(x)} \mathbb{1}[y_i = c]$. Here $\mathcal{N}_k(x)$ holds the k nearest fit images, and y_i is the label of fit image i. The indicator $\mathbb{1}[\cdot]$ is 1 if its condition holds and 0 otherwise. The class c is one of $C = 10$ classes.

Table 1 shows raw-pixel KNN with $k = 5$ and equal votes. The Euclidean run is ledger entry E1. The Manhattan and cosine runs come from a later logged rerun (`results/classical_reruns.json`). Manhattan is the most accurate, at 85.91 % test accuracy (+0.67 pp over Euclidean). Cosine is +0.39 pp over Euclidean, which is within noise. We did not test why Manhattan does better.

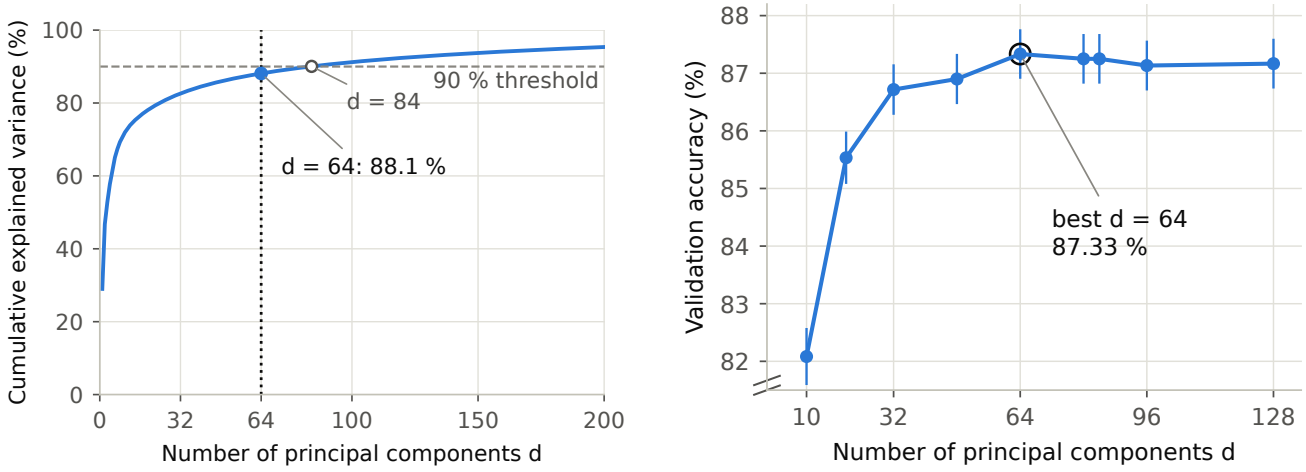
Table 1: KNN on raw 784-pixel inputs ($k = 5$, equal votes). Val and Test are accuracies in %. Macro F1 is the mean F1 score over the ten classes. Time is the median of 3 runs for 10,000 test queries on a 4-core Intel Xeon (2.10 GHz).

ID	Distance	Val (%)	Test (%)	Macro F1	Time (s)
E1	Euclidean	86.13	85.24	0.8518	4.64
—	Manhattan	86.97	85.91	0.8583	84.00
—	Cosine	86.18	85.63	0.8543	10.75

4.2 Reducing the Dimension with PCA

PCA [10] keeps the d directions in which the fit images vary most. We fit PCA on the 54,000 fit images only. Figure 2(a) shows that $d = 64$ components keep 88.1 % of the variance. The 90 % level is first reached at $d = 84$.

Figure 2(b) shows the validation accuracy of KNN ($k = 5$, Euclidean) on PCA features. The best value is at $d = 64$ (87.33 %). Beyond $d = 48$ the curve is flat within noise. On test data, PCA-64 with the same KNN gives 85.63 %. That is +0.39 pp over raw pixels, which is within noise.



(a) Cumulative explained variance.

(b) Validation accuracy versus d.

Figure 2: (a) Cumulative variance explained by PCA fitted on the 54,000 fit images. The dotted line marks $d = 64$ (88.1 %). The dashed line marks 90 %, first reached at $d = 84$. (b) Validation accuracy of KNN ($k = 5$, Euclidean) on PCA features (E2). The best value is at $d = 64$ (87.33 %), and the curve is flat beyond $d = 48$. Bars show ±1 SE.

4.3 Choosing k and Tuning KNN

A small k follows the fit images closely and is sensitive to noise. A large k gives smoother decisions but can mix neighboring classes. Figure 3 shows a sweep over $k = 1, 3, \ldots, 25$ on PCA-64 features (E4). Validation accuracy peaks at $k = 7$ (87.37 %). The values for $k = 5$, 7 and 9 are within noise.

We then tuned KNN on PCA-64 features with Optuna [9] for 40 trials (E3). The search covered k, the vote weights and the distance measure. The PCA dimension stayed fixed at 64. The best setting was $k = 5$, Manhattan distance, and votes weighted by inverse distance. Inverse-distance weights give closer neighbors more say. This model reaches 87.82 % validation and 86.43 % test accuracy (E3). That is +1.19 pp over raw-pixel KNN. On raw pixels, the Manhattan distance alone already gives 85.91 % (Table 1).

4.4 Query Time

Brute-force KNN compares each query with all 54,000 fit images. Its cost grows with the number of dimensions. Figure 4 shows the timing run in the ledger (E5). It timed all 6,000 validation queries in one call. Brute-force search took 2.98 s on raw pixels and 0.418 s on PCA-64 features, a 7.12× speed-up. The tree-based searches were slower than brute force.

We repeated the timing with a fixed protocol (`results/classical_reruns.json`). It used the 10,000 test queries, a 4-core Intel Xeon, one warm-up run and the median of 3 runs. Raw pixels took 4.64 s and PCA-64 took

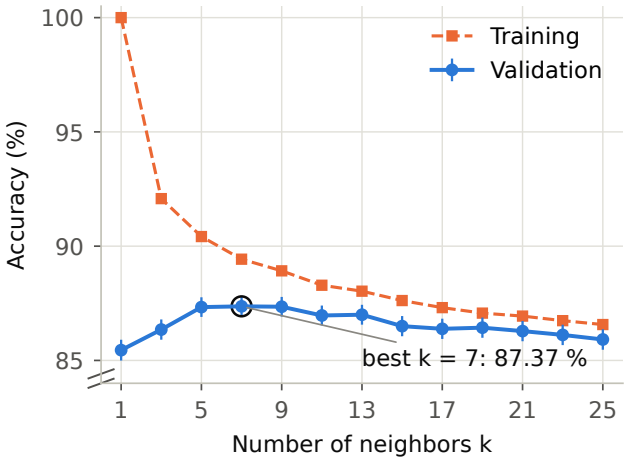


Figure 3: Training and validation accuracy of PCA-64 KNN (equal votes) for $k = 1$–25 (E4). Validation accuracy peaks at $k = 7$, but $k = 5$–9 are within noise. Training accuracy at $k = 1$ is 100 % because each training point is its own neighbor. Bars show ±1 SE.

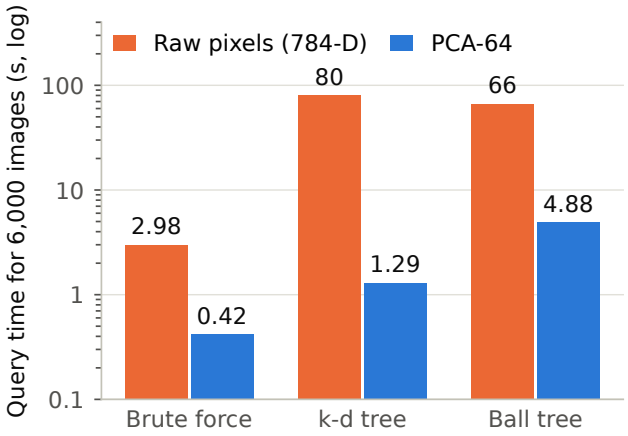


Figure 4: Query time of KNN for the 6,000 validation images on raw pixels and on PCA-64 features (E5, log scale, Euclidean distance). Brute force is the median of 3 runs. The k-d tree and ball tree were timed once and are slower than brute force.

1.49 s, a 3.1× speed-up. So the size of the speed-up depends on the hardware. The speed-up also does not hold for the tuned model. It uses the Manhattan distance and took 15.73 s, which is slower than raw-pixel Euclidean KNN.

5 Soft-KNN: Learning the Distance

5.1 Model

A majority vote does not change smoothly when the features change. So gradient descent cannot learn a better distance through it. Soft-KNN replaces the vote with a smooth score, as in NCA [4].

An embedding network f_θ maps an image x to a vector. Here θ are the network weights. We use the unit-length embedding $z = f_\theta(x)/\|f_\theta(x)\|_2 \in \mathbb{R}^{128}$. The simplest f_θ is a linear map $f_\theta(x) = Wx$, with a 128×784 matrix W.

A *memory bank* stores the embeddings $b_1, \ldots, b_N$ of all $N = 54{,}000$ fit images, with labels $y_1, \ldots, y_N$. For each class c, let $\mathcal{A}_c(x)$ be the m bank entries of class c that are closest to z. Note that m counts neighbors *per class* (the code calls it k). The mean distance to class c is

$$\bar{\delta}_c(x) = \frac{1}{m} \sum_{i \in \mathcal{A}_c(x)} \|z - b_i\|_2. \tag{2}$$

The model turns these distances into class probabilities with a temperature $\tau > 0$:

$$P_\tau(y = c \mid x) = \frac{\exp\bigl(-\bar{\delta}_c(x)/\tau\bigr)}{\sum_{c'=1}^{C} \exp\bigl(-\bar{\delta}_{c'}(x)/\tau\bigr)}. \tag{3}$$

Training minimizes the cross-entropy (CE) loss $-\log P_\tau(y \mid x)$ over the fit images.

What the model predicts. For every $\tau > 0$, the predicted class is $\arg\min_c \bar{\delta}_c(x)$. This is the class whose m nearest members are closest on average. It is a local-mean rule [11], not a majority vote. The temperature τ does not change the predicted class at test time. It only changes how confident the probabilities are. It also changes training, because the loss depends on these probabilities. The logged inference grids confirm this: in E19, the validation accuracy for each m is the same for $\tau = 0.1$, 0.3 and 0.5.

What is trained. Gradients flow only through the selected distances $\|z - b_i\|_2$ into the query embedding z. Choosing the m nearest entries is not differentiable. The bank is detached, so no gradient reaches the stored embeddings b_i.

Memory bank. Recomputing all 54,000 bank embeddings after every step would be slow. So we refresh the bank at fixed times, as in Wu et al. [7, 5]. The linear models refresh once per epoch (211 steps of 256 images). The MLP and convolutional models refresh every 107 steps. Unlike momentum contrast (MoCo) [12], there is no second, slowly updated encoder.

Self-match. Training images are also in the bank. So a training query finds its own (slightly older) embedding at a distance near zero. This inflates training accuracy and changes the training loss. Validation and test images are never in the bank. The code now has an opt-in fix (`--exclude-self`), but all logged numbers were produced without it.

5.2 Linear, MLP and Bank-Augmented Embeddings

The linear model (E6) maps 784 pixels to 128 numbers. Its matrix W starts from the PCA directions. We train it with Adam [13] (learning rate 10^{-3}, batch size 256, $m = 7$, $\tau = 0.5$). It reaches 88.22 % validation and 87.10 % test accuracy. That is +0.67 pp over the tuned classical KNN (E3).

A multilayer perceptron (MLP) with layers 784–256–128 did worse (E16, 85.70 % test). It used Adam with learning rate 3×10^{-4} and gradient clipping.

In E14 the bank also holds a shifted copy of each fit image (shift of up to 2 pixels). The queries stay unchanged. This gives 87.13 % test, +0.03 pp over E6, which is within noise. We still use this model, because the fusion gives it a large weight.

5.3 Temperature

We trained the linear model with six values of τ from 0.1 to 1.0 (E8). Figure 5 shows that validation accuracy falls as τ grows. The best tested value is $\tau = 0.1$ (86.53 %), at the edge of the grid. We did not test smaller values. These models were trained with flip, shift and erase augmentation and at most 10 epochs. They were all scored at $\tau = 0.5$. The scoring τ does not affect accuracy.

One possible reason for the trend: the embeddings have unit length, so each $\bar{\delta}_c$ lies in $[0, 2]$. At $\tau = 1$, no class can then get more than $1/(1 + 9e^{-2}) \approx 45$ % probability. So the loss stays high even for correct predictions. We did not test this explanation. The value chosen in E8 was not used for training later models. The CNN models were trained at $\tau = 0.5$ and scored at $\tau = 0.1$.

5.4 Optimizer Pilot

We compared three optimizers on the linear model (E9), with the same short, augmented schedule as E8. Stochastic gradient descent (SGD) used plain momentum 0.9 (not Nesterov) and learning rate 0.1. Adam [13] and RMSprop [14] used learning rate 10^{-3}. RMSprop scales each step by a running mean of squared gradients, which is a second moment.

Figure 6 shows the results. SGD was best: 83.85 % validation accuracy during training and 86.22 % after re-scoring. Adam reached 81.47 % and 82.50 %. RMSprop reached 80.83 % and 82.72 %. The learning rates differ by a factor of 100, and each optimizer ran once. So this is a small pilot, not a fair comparison. All later models still used Adam.

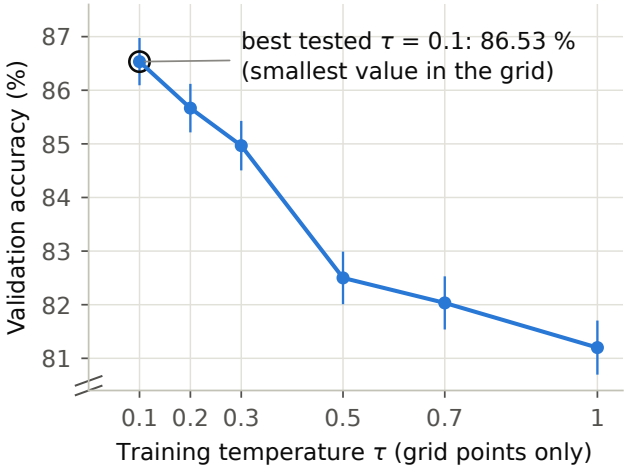


Figure 5: Validation accuracy of the linear Soft-KNN trained with each τ (E8). Only the six marked values were tested. Accuracy falls as τ grows. The best tested value, $\tau = 0.1$, is the smallest in the grid. Bars show ±1 SE.

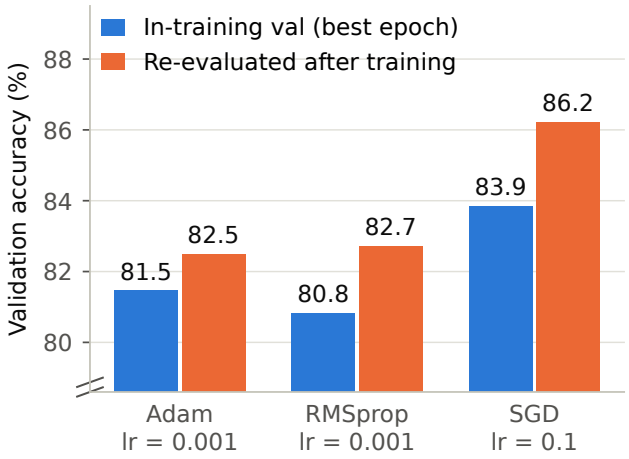


Figure 6: Optimizer pilot (E9) on the linear Soft-KNN with augmentation and at most 10 epochs. Blue: best validation accuracy during training. Orange: the returned model scored again with a fresh bank. Learning rates differ.

6 Convolutional Embedding, Ensembles and Fusion

6.1 Convolutional Embedding

A linear map ignores which pixels are neighbors. So we replace W with a small CNN. It has two blocks of 3×3 convolution, rectified linear unit (ReLU) and 2×2 max-pooling, with 32 and 64 channels. A linear layer then maps the $64 \times 7 \times 7$ features to 128 numbers, which we scale to unit length.

The first CNN run (E19, seed 42) used Adam with learning rate 3×10^{-4}, gradient clipping at 1.0 and batch size 256. It trained for the full 30 epochs. Early stopping, with a patience of 8 epochs, did not trigger. At

inference we use $m = 9$ and $\tau = 0.1$. It reaches 89.07 % validation and 87.63 % test accuracy, +0.53 pp over the linear model.

6.2 Seed Ensemble

We trained the same CNN with seeds $s \in \mathcal{S} = \{42, 7, 123\}$. These runs used automatic mixed precision (AMP), up to 35 epochs and a patience of 10 epochs. Seed 123 stopped after 14 epochs. The ensemble averages the class probabilities: $\bar{P}(y = c \mid x) = \frac{1}{3} \sum_{s \in \mathcal{S}} P^{(s)}(y = c \mid x)$. Here $P^{(s)}$ is the output of the model trained with seed s.

Figure 7 shows the three seeds and their ensemble (E21). The test accuracies are 87.53 %, 86.92 % and 87.57 %. Their mean is 87.34 % with a standard deviation of 0.36 pp. The ensemble reaches 88.06 %. That is +0.49 pp over the best seed, which is within noise, and +0.72 pp over the seed mean. The ensemble needs three networks at test time.

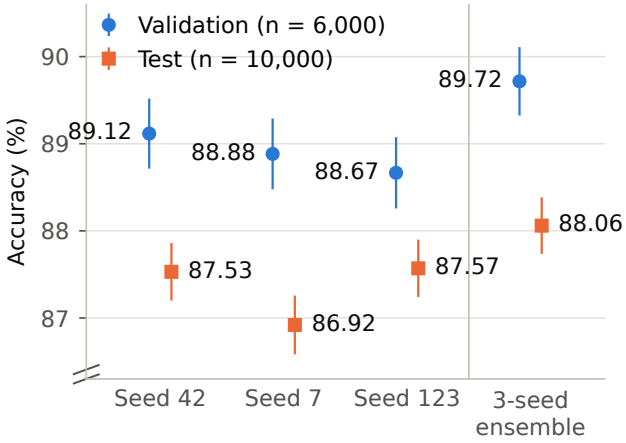


Figure 7: Validation and test accuracy of the three CNN seed models and of their ensemble (E21). Bars show ±1 SE. The seed spread on test is 0.65 pp.

6.3 Knowledge Distillation

Distillation [8] trains one student network to copy a teacher. Our teacher is the mean of three CNN seed models. For each fit image x_i, the teacher gives class probabilities $p^{\mathrm{T}}(x_i)$, computed with Soft-KNN at $m = 9$ and $\tau = 1$. The student is a new CNN with Soft-KNN scores $h_c(x) = -\bar{\delta}_c(x)/\tau$ (training $\tau = 0.5$). It minimizes

$$\mathcal{L} = (1 - \lambda)\,\mathrm{CE}\big(y, \mathrm{softmax}(h)\big) + \lambda\, T^2\, \mathrm{KL}\big(p^{\mathrm{T}} \,\big\|\, \mathrm{softmax}(h/T)\big). \tag{4}$$

Here KL is the Kullback–Leibler divergence, $\lambda = 0.5$ weights the two terms and $T = 1$ is the distillation temperature. With $T = 1$ the second term is simply $\mathrm{KL}(p^{\mathrm{T}} \| P_\tau)$.

The student and its three teacher models were trained on Kaggle. The teachers are a separate run of seeds 42, 7 and 123, with test accuracies 87.44 %, 87.25 % and 87.54 %. The student uses the same settings as the seed models.

The student (E23) reaches 90.10 % validation and 89.03 % test accuracy. That is +1.49 pp over its best teacher model and +0.97 pp over the E21 ensemble. This is a single run, and we do not claim a cause. A second Kaggle student, trained with five teacher seeds, reached 89.01 %. It is not used in any fusion.

Two caveats apply to the teacher targets. First, the teacher scored fit images that are in its own bank. Each image matched itself, so the targets partly encode the true labels. Second, at $\tau = 1$ the targets are far from one-hot. As shown above, no class can get more than about 45 %. So the targets act partly like smoothed labels.

6.4 Test-Time Augmentation and Fusion

Test-time augmentation (TTA) averages a model's predictions over six versions of each query: the original, a horizontal flip and four 1-pixel shifts. We apply it to the Soft-KNN sources in the last fusion.

The fusion combines J sources with weights: $P(y = c \mid x) = \sum_{j=1}^{J} w_j P_j(y = c \mid x)$, with $w_j \geq 0$ and $\sum_j w_j = 1$. We try all weight vectors on a grid with step 0.1. For the seven sources of E24 and E25, that is 8,008 vectors. We keep the vector with the best validation accuracy.

Table 2 lists the sources and weights. E22 fused six sources. E24 added the distilled student and reached 89.36 %. E25 added TTA and searched the weights again. It reached 89.51 % test and 90.85 % validation accuracy. The gain of E25 over E24 is +0.15 pp, which is within noise. Only three sources have nonzero weight in E25: the bank-augmented linear model (0.4), PCA-64 KNN (0.1) and the student (0.5).

Table 2: Fusion sources and the weights chosen on validation data (grid step 0.1). Soft-KNN sources use $\tau = 0.1$. TTA is used only in E25.

Source	Inference setting	E22	E24	E25
CNN seed 42	$m = 9$	0.2	0.2	0.0
CNN seed 7	$m = 9$	0.2	0.0	0.0
CNN seed 123	$m = 9$	0.1	0.0	0.0
Linear Soft-KNN, bank-augmented (E14)	$m = 5$	0.4	0.4	0.4
MLP Soft-KNN (E16)	$m = 9$	0.0	0.0	0.0
PCA-64 KNN (E3)	$k = 5$	0.1	0.1	0.1
Distilled student (E23)	$m = 9$	—	0.3	0.5
Validation accuracy (%)		90.27	90.62	90.85
Test accuracy (%)		88.73	89.36	89.51

7 Results

7.1 Main Results

Table 3 lists the main runs. The rows are grouped by method, not by the order in which they ran. The ledger also holds earlier fusions (E11, E15, E17) and diagnostic runs, which we leave out here.

Table 3: Main results on Fashion-MNIST. Val and Test are accuracies in % on 6,000 validation and 10,000 test images. "—" means the value was not logged. Rows without an ID, and the values marked †, come from the logged rerun of classical KNN.

ID	Method	Val (%)	Test (%)	Macro F1
E1	KNN, raw pixels, Euclidean, $k = 5$	86.13	85.24	0.8518
—	KNN, raw pixels, Manhattan, $k = 5$	86.97	85.91	0.8583
E2	KNN, PCA-64, Euclidean, $k = 5$	87.33	85.63†	0.8557†
E3	KNN, PCA-64, tuned with Optuna (Manhattan, $k = 5$, distance weights)	87.82	86.43	0.8644
E6	Soft-KNN, linear embedding	88.22	87.10	0.8698
E14	Soft-KNN, linear, bank with shifted copies	88.28	87.13	0.8703
E16	Soft-KNN, MLP embedding	86.92	85.70	—
E19	Soft-KNN, CNN embedding (seed 42)	89.07	87.63	—
E21	Ensemble of three CNN seed models	89.72	88.06	—
E23	Distilled CNN student	90.10	89.03	—
E22	Fusion of six sources	90.27	88.73	0.8858
E24	Fusion of seven sources	90.62	89.36	0.8923
E25	**Final fusion model**: seven sources with TTA	**90.85**	**89.51**	**0.8936**

The final fusion model (E25) is +4.27 pp above raw-pixel KNN (E1). It is +0.48 pp above the best single model, the distilled student (E23). That gap is within noise. The largest steps were tuning classical KNN (+1.19 pp, E1 to E3) and distillation (+0.97 pp over the E21 ensemble).

The gap between validation and test accuracy is 1.34 pp for E25. It is 1.44 pp for the single CNN model E19. So the fusion search did not add a large extra gap.

Figure 8 shows the errors of E25. Recall is highest for Bag (98.5 %), Trouser (97.9 %) and Sneaker (97.0 %). Shirt is lowest at 61.7 %. Of all shirts, 16 % are predicted as T-shirt/top. Most other errors are among Pullover, Coat and Shirt. The matrix is not stored in the ledger. Its counts were read from the E25 output figure, and their diagonal matches the logged 89.51 %.

7.2 Changes That Did Not Help

Table 4 lists changes that did not improve test accuracy.

Augmenting the training queries hurt in all three runs (E7, E13, E20). These runs were short, used one seed, and in E7 combined three changes. So they do not show that augmentation is harmful in general. One possible factor is self-match. Clean queries can match themselves in the bank, but augmented queries cannot.

TTA lowered the linear model from 87.10 % to 86.74 % (E10). This is within noise. On the augmented model of E7, TTA raised test accuracy from 81.68 % to 82.60 %. In E25, TTA was added together with a new weight search, so its own effect is unknown.

The second round of distillation (E26) used the E24 fusion as teacher, not the first student alone. Its teacher weights were 0.2 for CNN seed 42, 0.4 for the bank-augmented model, 0.1 for PCA-64 KNN and 0.3 for the

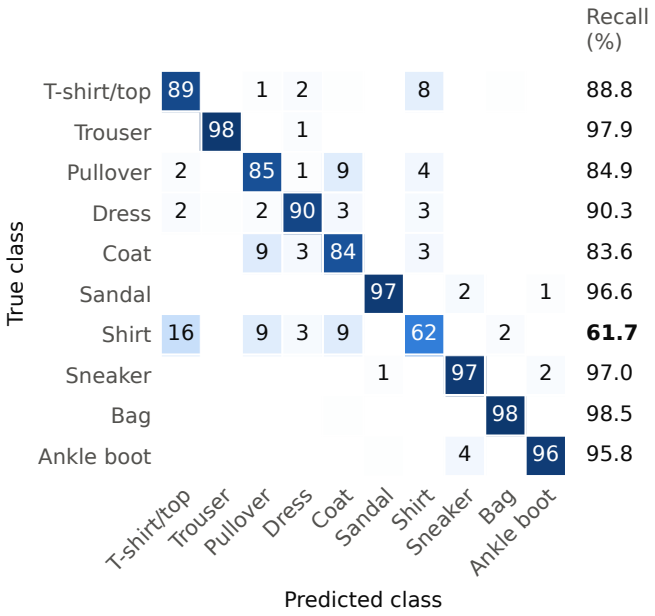


Figure 8: Row-normalized confusion matrix (%) of the final fusion model (E25) on the 10,000 test images. The last column is per-class recall. Recall is highest for Bag (98.5 %), Trouser (97.9 %) and Sneaker (97.0 %). Shirt is lowest (61.7 %), and 16 % of shirts are predicted as T-shirt/top.

Table 4: Changes that did not improve test accuracy. Δ is the difference in pp to the reference run. Gaps under about 0.6 pp are within noise.

ID	Change	Test (%)	Reference	Δ (pp)
E7	Flip, shift and erase on training queries (linear)	81.68	E6	−5.42
E13	Shift on training queries (linear)	83.66	E6	−3.44
E20	Shift on training queries (CNN)	84.62	E19	−3.01
E26	Second round of distillation	87.00	E23	−2.03
E16	MLP instead of linear embedding	85.70	E6	−1.40
E10	TTA on the linear model	86.74	E6	−0.36
E14	Shifted copies in the bank only	87.13	E6	+0.03

student. The teacher again scored fit images that were in its own banks. Its KNN part even returns the true label exactly, because each fit image is its own nearest neighbor at distance zero. The new student reached 87.00 %, which is 2.03 pp below the first student. The teacher changed in several ways at once, so we cannot say why.

8 Limitations

- **Self-match.** Training queries and teacher targets were computed with each image in its own bank. The opt-in `--exclude-self` fix was not used for any logged number.
- **Validation reuse.** The same validation images chose many settings, so validation accuracies are optimistic. Some inference settings ($m = 9$ for the CNN models, $m = 5$ for the bank-augmented model) were fixed by hand.
- **Stale bank.** Early stopping scored validation images against the bank from the last refresh, not against fresh embeddings.
- **Noise.** Most results are single runs. We ran no significance tests, and the CNN seeds alone differ by up to 0.65 pp.
- **Missing controls.** We did not test a hard KNN vote on the learned embedding. We did not train the same CNN with a standard classification layer. So we cannot say whether the neighbor-based output helps.
- **Scope.** We used one dataset. The final accuracy is in the range of the classical baselines reported with the dataset [3], such as support vector machines.
- **Mixed environments.** Some models were trained on Kaggle, and AMP makes GPU runs not exactly repeatable. The same seeds gave different accuracies in two runs (`results/conv_seeds.json` and `results/kaggle_conv_seeds.json`).
- **Notebook.** The Colab notebook is a simplified demo. It does not reproduce the reported numbers (Section 10).

9 Conclusion and Future Work

We raised test accuracy on Fashion-MNIST from 85.24 % (raw-pixel KNN) to 89.51 % (the final fusion model, E25). Tuning classical KNN on PCA features gave 86.43 %. Learning the distance with Soft-KNN gave 87.10 % with a linear embedding and 87.63 % with a CNN. Ensembling, distillation and fusion gave the rest. Every model still decides from distances to stored training images. However, the final system depends on a trained neural embedding and on several combined models.

Future work should rerun the learned models with self-match excluded and with several seeds. It should add the missing controls, use a separate split for the fusion weights, and test other embedding sizes.

10 Reproducibility

The code, the ledger and the figures are in the GitHub repository linked on the title page. All reported numbers come from the following commands:

- `python -m src.experiments --scale full --stages E1,E2,...` runs the ledger stages and appends to `results/results.json`.
- `python scripts/train_conv_seed.py --seed 42` and `python scripts/train_distill.py` train the seed models and the student.
- `python scripts/rerun_classical.py` repeats the classical KNN runs and timings.
- `python scripts/make_paper_figures.py` draws the figures, and `bash docs/build_papers.sh` builds this report.

Trained models are not stored in the repository. So the stages from E10 on need the models to be trained again. The fusion stages use whichever trained models exist, so a rerun may use other sources than the logged run.

The Colab notebook is a short teaching demo. It runs classical KNN on a subset of the images and a small CNN with learned class prototypes. It has no memory bank, ensemble, distillation or fusion, and it does not reproduce the numbers in this report.

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